

# Nitish Joshi

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## Education

- 2016-Present **Bachelor of Technology**, *Indian Institute of Technology*, Bombay, .  
Major: Computer Science and Engineering (with honors)  
CGPA: 9.26/10.00
- 2014-2016 **Intermediate/+2**, *Late P.B. Jog Junior College*, Pune, *Percentage* - 92.46.
- 2014 **Matriculation**, *Vikhe Patil Memorial School*, Pune, *CGPA*: 10/10.

## Interests

Natural Language Processing, Deep Learning, Reinforcement Learning

## Research Projects

May '18 - **Multi-hop Reading Comprehension**.

Present Prof. Mohit Bansal: University of North Carolina, Chapel Hill

- Proposed novel deep learning algorithms for multiple-hop reasoning based question answering systems
- Explored various ideas related to **attention mechanisms** from state-of-the-art techniques in machine comprehension such as Bidirectional Attention Flow Network and Hierarchical Attention Networks
- Implemented **policy gradient** methods for optimizing parameters in case of a non-differentiable loss
- Experimented with various techniques in **Reinforcement Learning** specifically related to the problem of **credit assignment** and implemented the same in python using **Tensorflow**

July '18 - **Automatic question generation**.

Present Prof. Preethi Jyothi, IIT Bombay

- Working on building novel models for automatic question generation for low resource languages
- Building unsupervised and semi-supervised models by leveraging information from data rich languages like English and implementing the same in **PyTorch**

Spring 2018 **Accent Adaptation in Speech Recognition**.

Prof. Preethi Jyothi, IIT Bombay

- Built an end-to-end speech recognition system which can learn domain invariant features
- Incorporated ideas from the field of **Domain Adaptation** in Computer Vision and NLP such as Domain Adversarial Training of Neural Networks and Multinomial Adversarial Networks
- Explored **Attention-based** models and **CTC** models in automatic speech recognition systems

## Miscellaneous Projects

Summer 2017 **Automated detection and grading of prostate cancer**.

Prof. Amit Sethi, IIT Bombay

- Learnt basic concepts of **Multilayer Perceptrons**, **Convolutional Neural Networks** and **Residual Neural Networks** and implemented the same on standard datasets using **Tensorflow**
- Reviewed literature related to existing methods used for automated detection, grading and recurrence detection of prostate cancer
- Studied methods to grade cancer in H&E stained needle biopsies of prostate tissues using gleason score

#### Autumn 2017 **Document based Similarity Analysis.**

Course Project - CS251: Software Systems Lab

- Building a comprehensive 'Document Similarity' system primarily aimed at detecting duplicate content
- Implementing basic algorithms like **Fingerprint Algorithm** and **Jaccard Similarity** as well as clustering algorithms for multiple set of documents

#### Spring 2018 **Railway Controller.**

Course Project - CS254: Digital Logic Design Lab

- Designed a communicating railway signal controller in VHDL and tested it by mapping it to a FPGA board
- Used **FPGAlink** for encrypted communication with host and **UART** for inter-controller communication

## Academic Achievements

- 2017 Received **Institute Academic Award** for academic excellence, awarded to the top 15 students in a batch of 900 students at IIT Bombay
- 2017 Awarded **AP Grade** for exceptional performance in Linear Algebra and Chemistry Lab
- 2016 An **All-India rank of 176** in JEE-Advanced among 150 thousand candidates
- 2016 An **All-India rank of 139** among a total of one million applicants in JEE-Main exam
- 2016 Received **Silver** medal (given to about top 25%) at **USA Physics Olympiad**
- 2016 **National top 1%** in NSEP (Physics Olympiad) and NSEC (Chemistry Olympiad) in India
- 2013,2014 Placed **8th and 26th in State** in Regional Mathematics Olympiad, and secured merit certificate in **Indian National Mathematics Olympiad**

## Technical Skills

- Languages C++, Python, Java
- Other tools MATLAB,  $\text{\LaTeX}$ , Git, Gnuplot
- Libraries TensorFlow, OpenCV

## Key Courses

- Computer Science Automata Theory\*\*, Implementation of Programming Languages\*\*, Artificial Intelligence\*\*, Computer Architecture\*, Operating Systems\*, Data Structures and Algorithms, Natural Language Processing with Deep Learning (online), Software Systems, Discrete Structures, Data Analysis and Interpretation, Machine Learning (online), Design and Analysis of Algorithms, Logic for CS, Computer Networks
- Mathematics Numerical Analysis\*\*, Linear Algebra, Differential Equations, Calculus
- Miscellaneous Quantum Physics and Application, Biology, Introduction to Electrical Systems, Introduction to Electronics

\* to be completed by December '18

\* to be completed by April '19

## Extra-curricular activities

- Gave a talk on the paper 'Natural Language Decathlon: Multitask Learning as Question Answering' as part of the UNC - Natural Language Processing reading group
- Built a remote controlled robot using ATtiny microcontroller for XLR8, an obstacle race course organized by Robotics Club, IIT Bombay
- Played 'Tabla', an Indian percussion instrument for six years with various solo and group performances
- Completed a year long course in badminton as a part of the National Sports Organization.
- Participated in code.fun.do Hackathon organized by Microsoft